

37 — EM ALGORITHM FOR GAUSSIAN MIXTURE MODELS

Aim

To implement the Expectation-Maximization algorithm using a Gaussian Mixture Model and identify hidden clusters in a dataset.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- NumPy
- scikit-learn

Algorithm + Procedure

1. Generate clustered data using `make_blobs()`.
2. Create a Gaussian Mixture Model with three components.
3. Initialize the model parameters.
4. Perform the E-step to calculate component responsibilities.
5. Perform the M-step to update model parameters.
6. Repeat until convergence.
7. Predict the cluster labels.
8. Display the convergence status and estimated means.

Code

```
import numpy as np

from sklearn.mixture import GaussianMixture

from sklearn.datasets import make_blobs

X, y_true = make_blobs(
    n_samples=300,
    centers=3,
    cluster_std=0.7,
    random_state=1
)

gmm = GaussianMixture(
```

```
n_components=3,  
random_state=1  
)  
gmm.fit(X)  
labels = gmm.predict(X)  
print("Converged:", gmm.converged_)  
print("Iterations:", gmm.n_iter_)  
print("Estimated means:")  
print(np.round(gmm.means_, 2))
```

Output

```
Converged: True  
Iterations: 2  
Estimated means:  
[[ -1.52  4.41]  
 [-10.04 -3.93]  
 [ -7.04 -8.06]]
```

Result

The EM algorithm was successfully implemented using a Gaussian Mixture Model. The model converged and estimated the parameters of the hidden Gaussian clusters.